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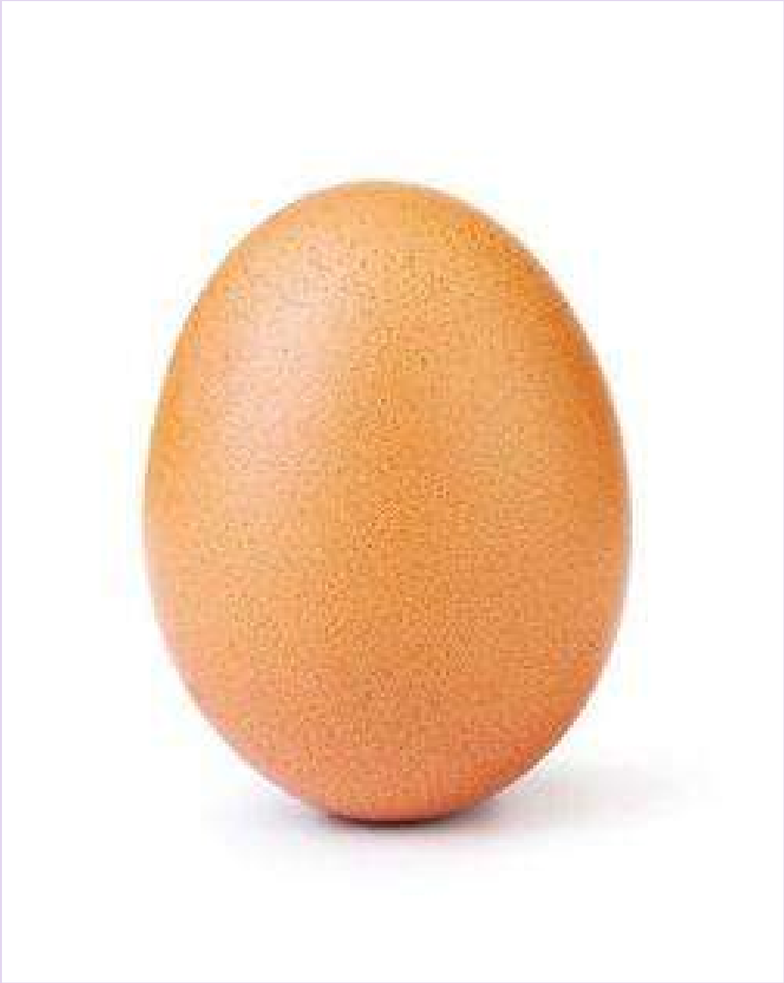
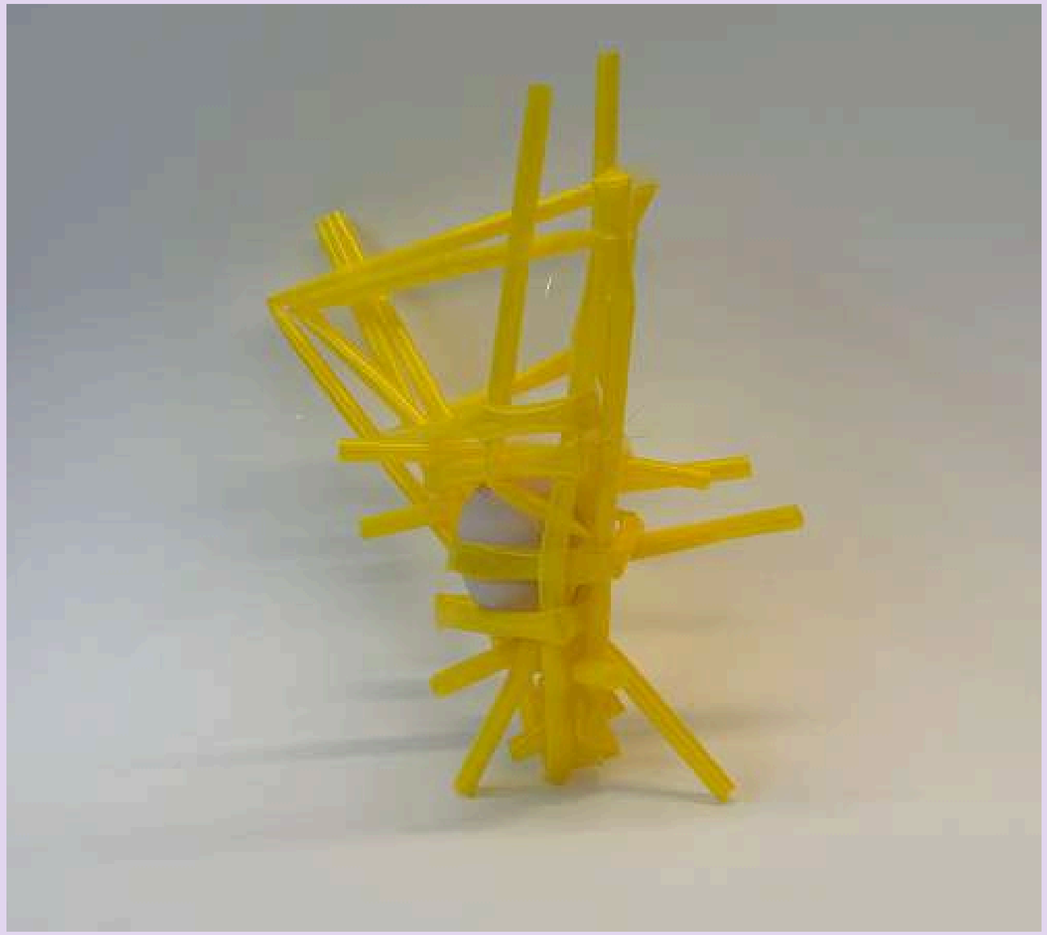
Egg Drop

Milestones

It started with an egg...

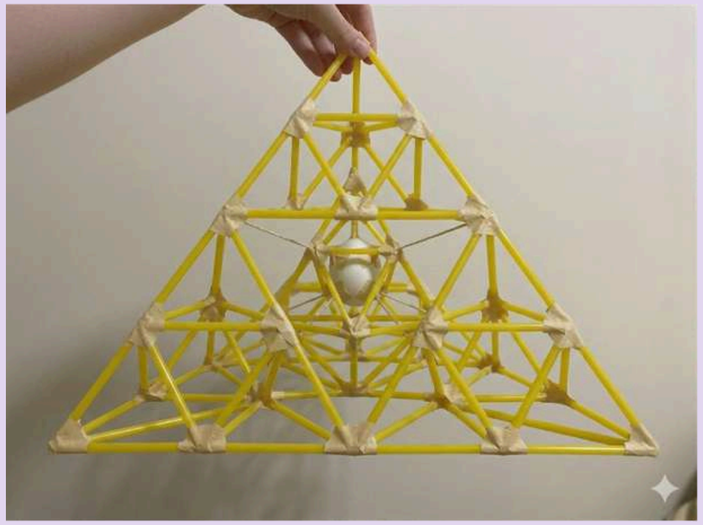
“Don’t let it crack,” they said.
So I went to Şok and there were no straws left.
Failing that, I went to Migros. They did in fact have a few hundred straws left.
The toughest part of the first prototype is that **we couldn’t use glue or tape;** we could only use string. **It's really hard to keep a plastic structure together with string.**

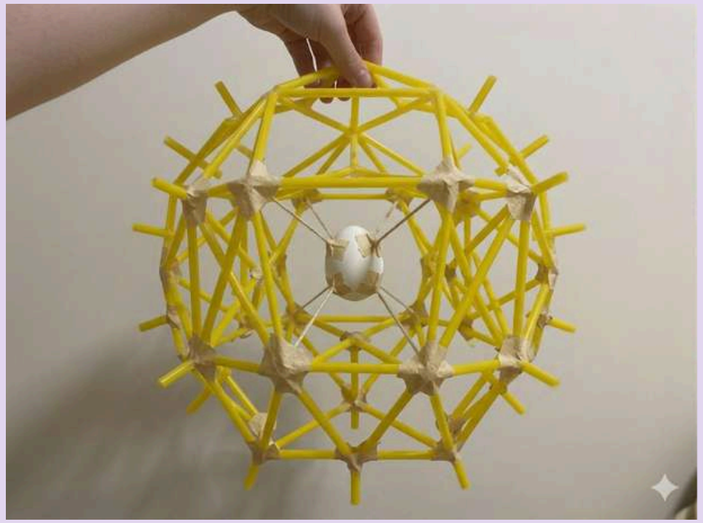


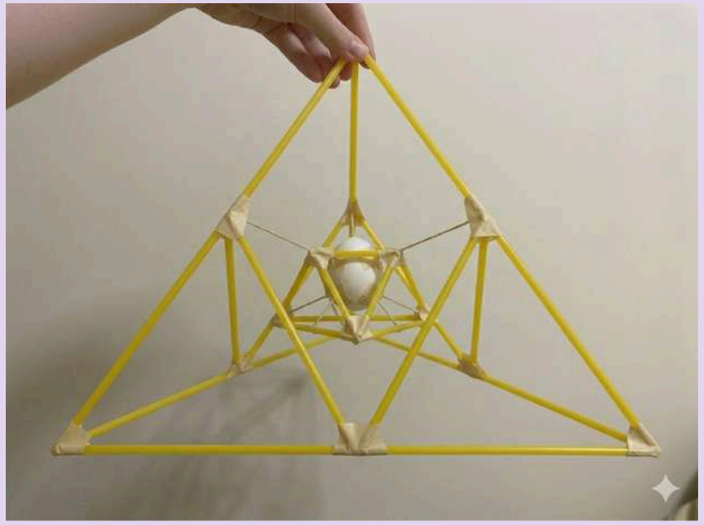


“I talked to my family, and my brother, who is an industrial engineer, told me to make a triangle because it’s a structurally sound shape. Just look at the pyramids. I really just took a few pieces of straw and made a triangle out of them. Then I added loose pieces around as padding for the egg so it didn’t immediately break if it fell on its side. All in all, it did its job. It survived the 1.5-meter drop, no cracks in sight. It, however, failed the 3-meter fall, which was never the assignment, so I can’t be held to that.”

With the new limits, 20 straws max and 20 cm max, I faced a new challenge. I gave ai these limits and it gave me:





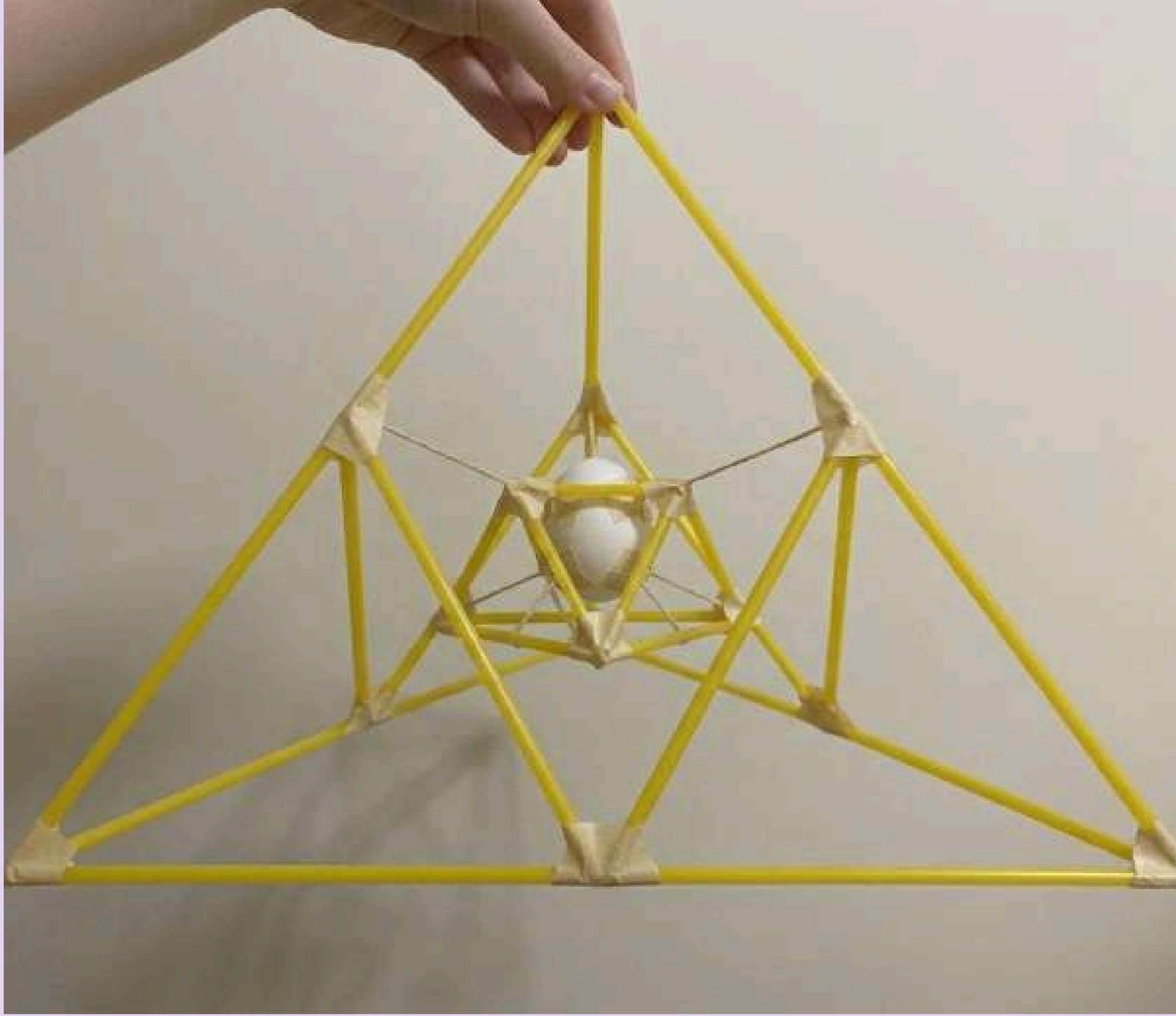


AI response

None of these were doable with these limits. I rejected these.

AI just failed to understand the limits, and if it learned to abide by one of the rules, it immediately forgot the others. I learned that I have to be incredibly specific, or it doesn’t do a good job.

Gemini did create a bunch of images, but none of them were useful to me, and the whole process felt really frustrating. I accepted the general geometry ideas.



Prototype 2

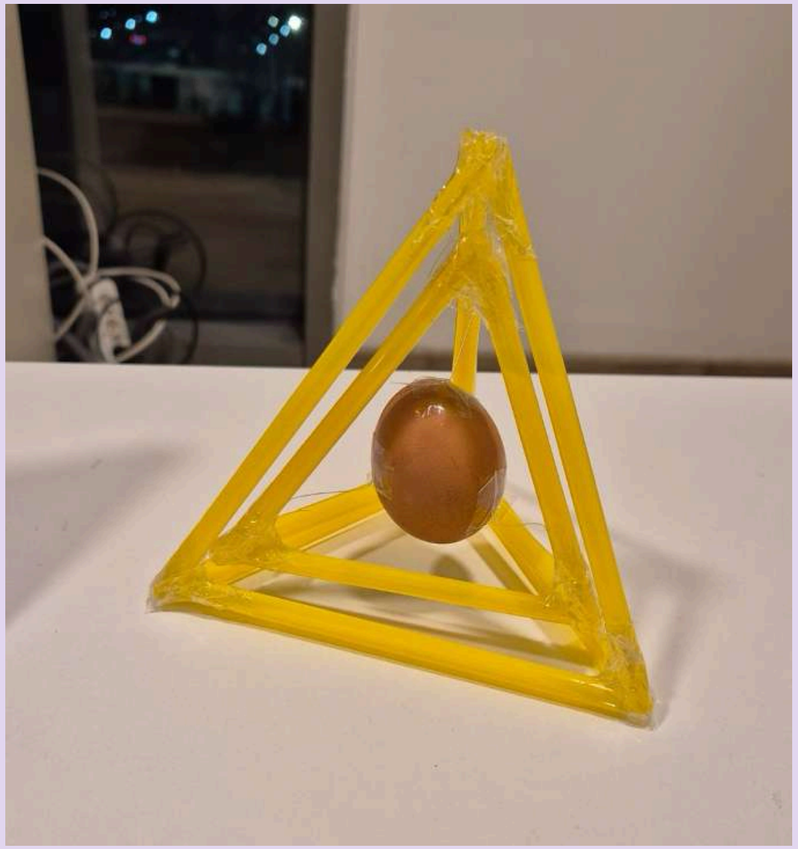
Partnered with AI

AI suggested i have more **“crumple zones”** so less tension reaches the egg. So I tried this and it didn’t reach the 3 meters. **The straws over the egg put too much tension on the egg.** AI suggested I use strings to **suspend the egg** instead. I took this idea.

Prototype 3

Partnered with AI

In this one the **strings weren’t secure enough**. The egg made contact with the floor instead of being suspended between the straws. AI didn’t have a solution for this other than to **make it bigger**, which I couldn’t do because of the limit of 15 cm. Gemini said that the space between the triangles needed to be larger. **I abandoned this idea.**



Final Prototype

At this point, **I stopped using AI.** I made a triangle again so it's secure; I added padding on the sides so the egg doesn’t crack if it lands on its side. **This too made it 1.5 meters but not 3.**

I used AI to expand on my knowledge and improve my ideas. I don’t think this was very successful.

In conclusion, the egg drops using AI weren’t any more successful than the first version where I didn’t use AI.

Gemini really struggled to learn the limits of the project and gave me undoable ideas.